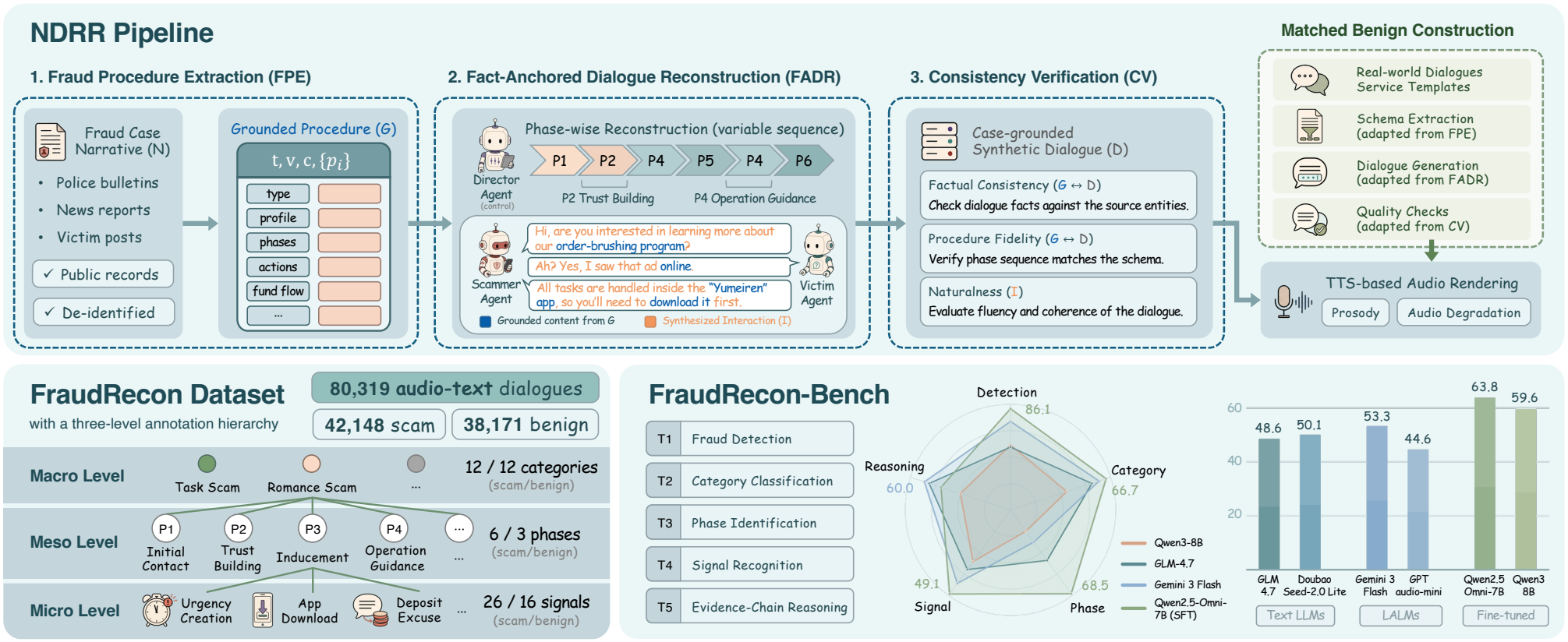

NDRR Pipeline
1. Fraud Procedure Extraction (FPE)
Fraud Case Narrative (N)
• Police bulletins
• News reports
• Victim posts
✓ Public records
✓ De-identified
Grounded Procedure (G)
t, v, c, {p_i}
type
profile
phases
actions
fund flow
...
2. Fact-Anchored Dialogue Reconstruction (FADR)
Phase-wise Reconstruction (variable sequence)
Director Agent (control)
P1 P2 P4 P5 P4 P6
P2 Trust Building
P4 Operation Guidance
Scammer Agent
Hi, are you interested in learning more about our order-brushing program?
Ah? Yes, I saw that ad online.
All tasks are handled inside the "Yumeiren" app, so you'll need to download it first.
Victim Agent
Grounded content from G
Synthesized Interaction (I)
3. Consistency Verification (CV)
Case-grounded Synthetic Dialogue (D)
Factual Consistency (G ↔ D)
Check dialogue facts against the source entities.
Procedure Fidelity (G ↔ D)
Verify phase sequence matches the schema.
Naturalness (I)
Evaluate fluency and coherence of the dialogue.
Matched Benign Construction
Real-world Dialogues Service Templates
Schema Extraction (adapted from FPE)
Dialogue Generation (adapted from FADR)
Quality Checks (adapted from CV)
TTS-based Audio Rendering
Prosody
Audio Degradation
FraudRecon Dataset
with a three-level annotation hierarchy
80,319 audio-text dialogues
42,148 scam
38,171 benign
Macro Level
Task Scam
Romance Scam
...
12 / 12 categories
(scam/benign)
Meso Level
P1 Initial Contact
P2 Trust Building
P3 Inducement
P4 Operation Guidance
...
6 / 3 phases
(scam/benign)
Micro Level
Urgency Creation
App Download
Deposit Excuse
...
26 / 16 signals
(scam/benign)
FraudRecon-Bench
T1 Fraud Detection
T2 Category Classification
T3 Phase Identification
T4 Signal Recognition
T5 Evidence-Chain Reasoning
Detection
86.1
Category
66.7
Phase
68.5
Signal
49.1
Reasoning
60.0
Qwen3-8B
GLM-4.7
Gemini 3 Flash
Qwen2.5-Omni-7B (SFT)
60
40
20
48.6
50.1
53.3
44.6
63.8
59.6
GLM 4.7
Doubao Seed-2.0 Lite
Gemini 3 Flash
GPT audio-mini
Qwen2.5 Omni-7B
Qwen3 8B
Text LLMs
LALMs
Fine-tuned